

Report On
CLASSIFICATION AND PREDICTION TECHNIQUES

Department of INFORMATION TECHNOLOGY
by

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Vulnerability Assessment & Attack Strategy

Q1. > An e-commerce website was attacked during a festive sale. Later investigation revealed weak password policies and outdated software.

Analyze the planning mistakes that allowed the attack to succeed. Explain how vulnerability identification and timing strategy played a role.

Ans) During the festive sale, the e-commerce website became vulnerable due to weak security planning. The main mistakes are explained below:

1) Poor password policy

- The company allowed weak passwords.
- This made it easy for attackers to guess or crack user/admin accounts.
- Under Cyber law, organizations are responsible for protecting user data.

2) Outdated software and Systems

- The website was running old versions of software.
- Such software contains known security loopholes.
- Attackers exploit these publicly known vulnerabilities.

3) Lack of Regular Security Audits

- The company did not perform regular vulnerability testing.
- Weak points in the system remain undetected.
- This violates ethical responsibility to ensure digital safety.

4) Poor risk Assessment during high traffic.

- Festive sales attract heavy user traffic.
- The company failed to strengthen security during this period.
- Attackers took advantage of increased system load.

5) Weak Access Control

- Admin panels and databases were not properly protected.

- Unauthorized access became possible.

Role of Vulnerability Identification —

- Attackers scan websites to find weak passwords and outdated systems.
- They use tools to detect open ports, unpatched software, and misconfigurations.
- Once vulnerabilities are identified, attackers prepare targeted attacks.

Role of timing strategy —

- Attackers chose the festive season because:
 - Traffic was high
 - Monitoring was weaker
 - Technical teams were overloaded
- Any unusual activity was less likely to be noticed.

Q2) A company experienced a DDoS attack during a public product launch event. Analyze why attackers might choose specific timing and attack methods. Compare the impact of poor preparation versus strong cybersecurity planning.

Ans) A DDoS (Distributed denial of service) attack aims to overload a website and make it unavailable. Launch events are ideal targets for such attacks.

1) Reason for choosing specific timing.

Attackers selected the product launch because:

- Maximum number of users visit the website.
- Server load is already high.
- Even a small attack can cause major disruption.
- Media and public attention increases impact.

This increases financial and reputational damage.

2) Choice of Attack Method

Attackers use DDoS attacks because:

- They are easy to execute using botnets.
- Attack traffic comes from multiple sources.
- It is difficult to trace attackers.

- It quickly brings down servers.
- Common methods include:
- Traffic flooding
 - HTTP request attacks
 - Botnet-based attacks.

3) Impact of Poor preparation

If cybersecurity planning is weak:

- No backup servers
- No traffic filtering system
- No emergency response plan
- No monitoring tools.

Then:

- Website crashes
- Customers lose trust
- Financial loss occurs
- Legal issues may arise.

4) Impact of Strong Cybersecurity Planning

With proper planning:

- DDoS protection systems are installed
- Firewalls and load balancers are used.
- Backup servers are active
- Security teams are alert.

Results:

- Attack is controlled quickly
- Services remain available
- Customers confidence is maintained
- Legal compliance is ensured.

5) Cyber law and Ethical perspective

- DDoS attacks are illegal under cyber laws.
- Attackers violate digital ethics and user rights.
- Companies must ethically protect digital infrastructure.
- Failure may lead to illegal liability.